Impact of Iron Availability on the Motility of *Escherichia coli* CFT073

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Introduction

Urinary tract infections (UTIs) are the second most common type of bacterial infection, affecting primarily women. Approximately 50% of women will contract a UTI at some point in their lives. UTIs are commonly caused by uropathogenic *Escherichia coli* (UPEC) and involve bacterial motility as a virulence factor. Swimming and swarming are flagella-driven types of motility, with swarming being collective movement on solid surfaces. TonB, part of the TonB-ExbBD complex, energizes iron uptake via outer membrane receptors. We are looking at 2 isogenic mutants which have different iron acquisition capabilities. CFT073 ΔtonB has low intracellular iron since the import of iron is compromised. CFT073 Δfur has high intracellular iron; the transcriptional repressor in control of iron import is lacking, allowing excess iron import via iron acquisitions system like TonB. We hypothesized that iron limitation influences swarming motility in UPEC. Understanding the relationship between iron limitation and the degree of swarming motility may allow us to invent novel treatment methods for UTIs.

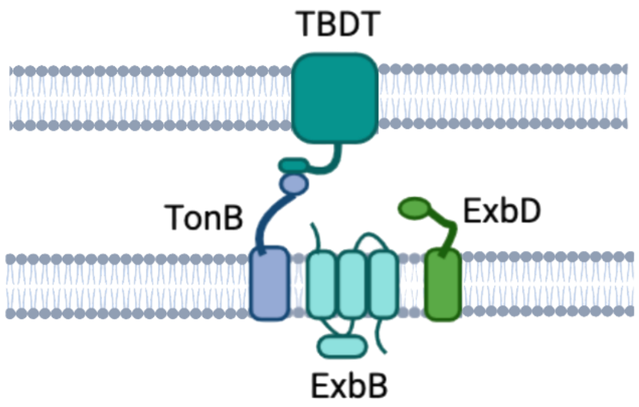

Figure 1. Important membrane protein complexes are depicted: the TonB-ExbBD complex. Created using BioRender.com

Methods

- *E. coli* CFT073 WT, *ΔtonB,* & *Δfur* were incubated overnight in TSB at 37°C.
- Swarming agar plates were poured and left covered at room temperature overnight.
- Swimming agar plates were poured 30 minutes prior to use
- Appropriate concentrations of Fe and DIP were added to plates before pouring
- OD was standardized (OD600=1), and 5µL of cells were inoculated on Eiken agar (swarming) plates. Swimming plates were stab inoculated.
- Swarming plates were incubated at 37 °C for 24 hours. Swimming plates were incubated at 30 °C for 18 hours
- Images of plates were taken, and swarming areas were measured using ImageJ software.

Figure 2. While individual WT *E. coli* expresses swimming motility in liquid media, multicellular swarming motility is expressed in solid. This image was created with Biorender.com.

Results

To uncover the role of iron availability in motility, we tested different strains of uropathogenic *E. coli* with differing concentrations of added iron and dipyridyl (an iron chelator) and measured the subsequent swarming areas.

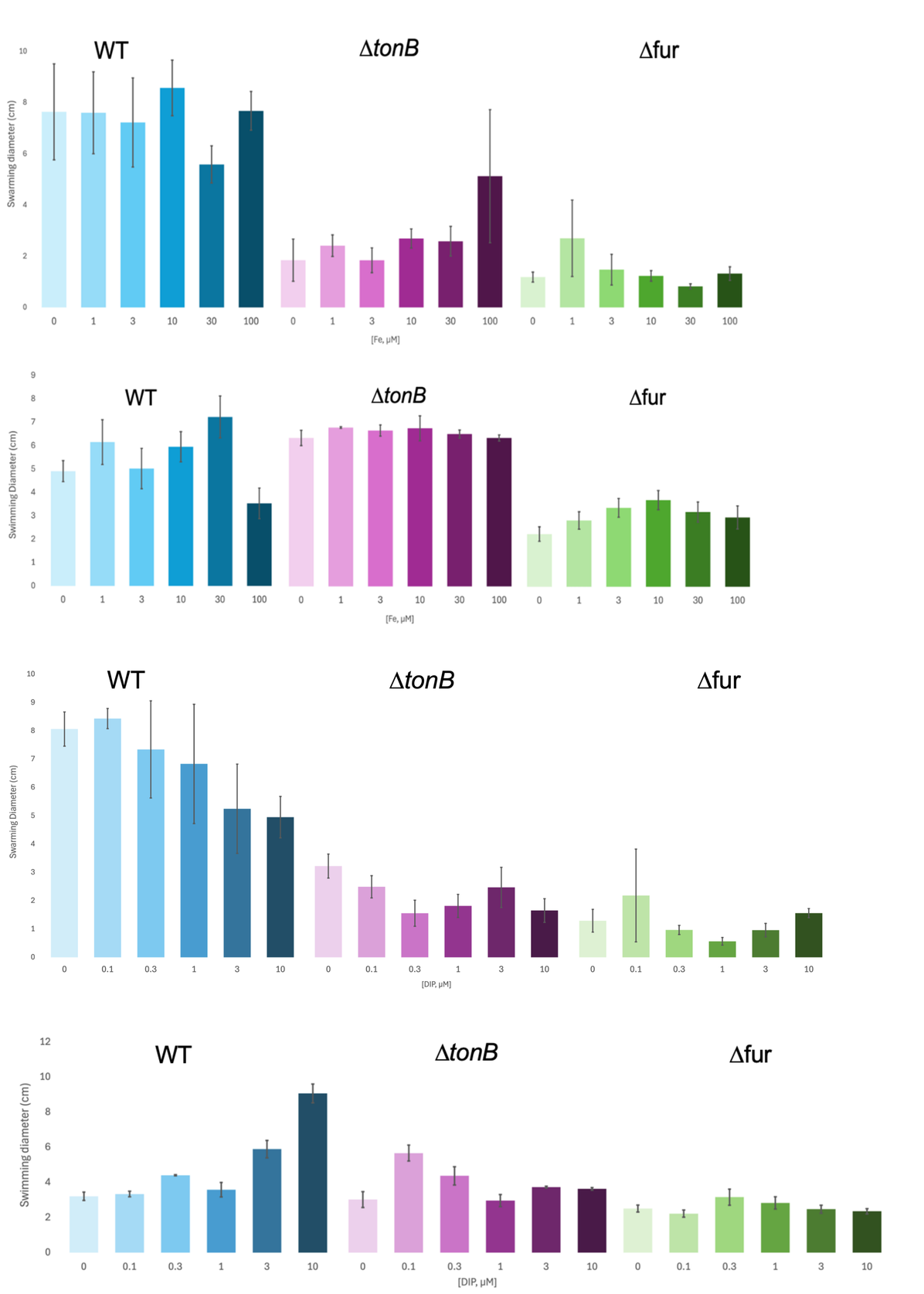

Figure 3. These graphs show the effect of both added iron (Fe) and chelated iron (DIP) on swimming and swarming motility of three UPEC isogenic mutants.

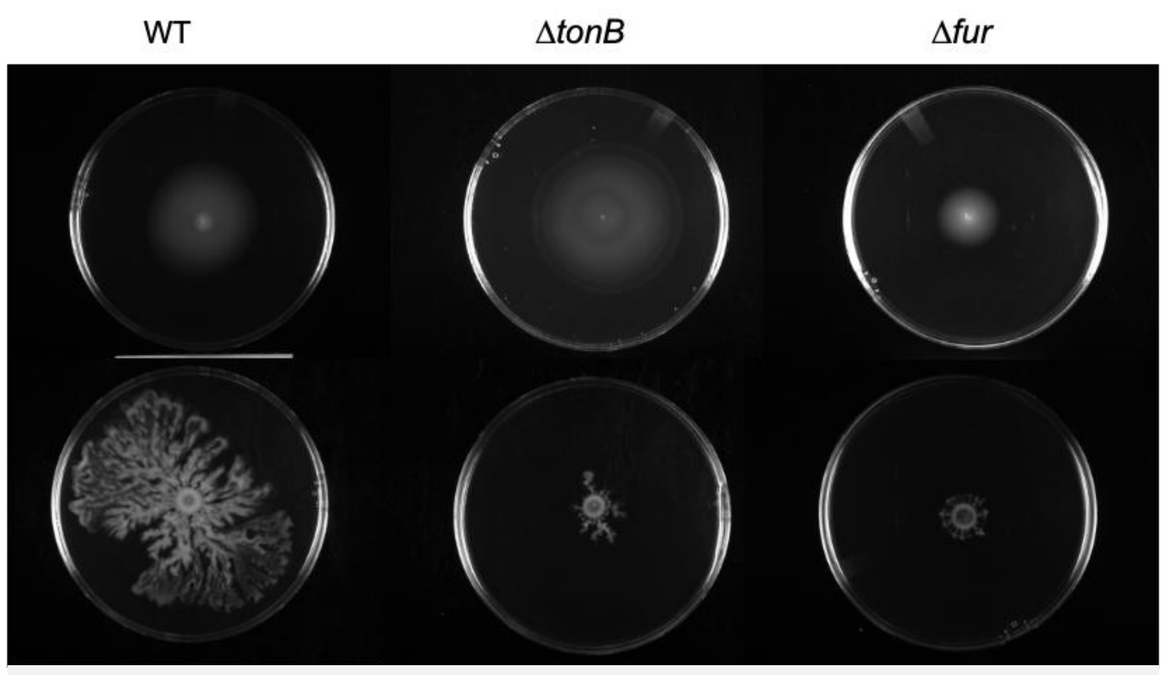

Figure 4. Example images of *E. coli* CFT073 WT, *ΔtonB,* and *Δfur* on swimming (top) and swarming (bottom) media. Swimming plates are after 18hr incubation. Swarming plates are after 24hr incubation.

Conclusion

We found that added concentrations of iron tend to decrease the swarming area of wild type *E. coli* CFT073, and that increased concentrations of an iron chelator, dipyridyl, tends to increase swarming area. However, DIP arrests all growth at concentrations greater than 100 µM. The *ΔtonB* mutant showed no change in phenotype across all tested conditions. This led us to the conclusion that iron availability is most likely not the cause of the mutant's inability to swarm. The patterns observed support the hypothesis that iron limitation increases swimming motility. Currently, we are testing how iron availability impacts attachment, via biofilm formation. Attachment is an important virulence factor in urinary tract infections because it is required to attach to the walls of the bladder. We plan to further explore the impacts of iron availability on the biofilm formation and bacterial growth of three CFT073 mutants.

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References

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